sysMonitorHistory

Service Description

Service ID: *"sysMonitorHistory"*

Abstract
This document describes the `sysMonitorHistory` service, version 1.0.0, produced by the SysMonitor support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `sysMonitorHistory` service, which provides query of the stored sysMonitor observations of every monitored system. It is produced by the SysMonitor system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

New in this workspace (SysMonitor-History-HTTP-IDD-5.2.2).

2.2 How This Service Is Meant to Be Used

LocalCloudManager and the PlantDescriptionEngine read the systems, the latest observation per type and the history of a system.

2.3 Important Delimitations and Dependencies

Every operation needs a ConsumerAuthorization grant (`sysMonitorHistory`, scope = operation; cached 30 s). History is kept `SYSMONITOR_RETENTION_DAYS` (30) days.

3 Service Interface

This section describes the interfaces to the `sysMonitorHistory` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

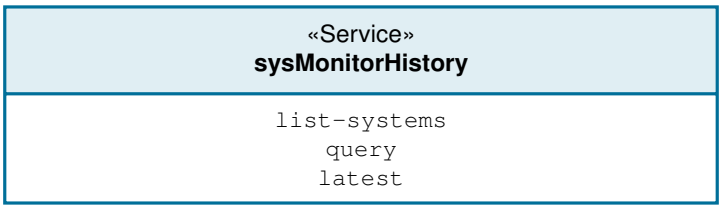


Figure 1: SysML-style block of the `sysMonitorHistory` service and its operations.

The following interface operations are available.

3.1 operation list-systems () : SystemList

Returns the systems with stored observations.

3.2 operation query (Name) : ObservationList

Returns observations of a system, optionally by type and period (`count` 1..1000, default 100).

3.3 operation latest (Name) : Latest

Returns the latest observation per type of a system (404 if none).

4 Information Model

Here, all data objects that can be part of the `sysMonitorHistory` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.5, which are used to represent things like hashes and identifiers.

4.1 struct Latest

Field	Type	Description
systemName	Name	The system.
latest	Map<String, Object>	Latest observation per type.

4.2 struct ObservationList

Field	Type	Description
systemName	Name	The system.
count	Number	Number of observations.
observations	List<StoredObservation>	Newest first.

4.3 struct StoredObservation

Field	Type	Description
schemaVersion	String	`1.0`.
observationId	Identifier	Identifier (publisher's, or a SHA-256 of routing key and content).
providerSystem	String	Publishing system.
serviceInstance	String	Service instance.
observationType	String	Observation type.
sourceTime	DateTime	Publisher time, UTC.
receivedTime	DateTime	Time SysMonitor received it.
publishedTime	DateTime	Time it was forwarded on `sysMonitorData`.
value	Map<String, Object>	Observed value.
unit	String	Unit.
quality	String	Quality.
pipeline	Map<String, Object>	Intake details (routing key, exchange).

4.4 struct SystemList

Field	Type	Description
systems	List<String>	Systems with stored observations.
retentionDays	String	History retention in days.

4.5 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.

5 References

[1] J. Delsing, “SysMonitor – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-SysMonitor-java-spring/documentation/v5.3.0/SysMonitor-v5.3.0-SysD.pdf.

[2] ——, “sysMonitorHistory HTTP/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.3.0, 2026, sysMonitor-v5.3.0-sysMonitorHistory-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	